Version 7.4.7

PVsyst - Simulation report

Grid-Connected System

Project: SPV5_Kolkata_5MWp

Variant: Bifacial 20deg, PITCH 8.0m, 3D sheds, soiling 3pct

Sheds, single array

System power: 4995 kWp

Kolkata - India

Author

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Project summary

Geographical Site
Kolkata
India

Situation

Latitude	22.56 °N
Longitude	88.36 °E
Altitude	11 m
Time zone	UTC+5.5

Project settings

Albedo	0.20

Weather data
Kolkata
Meteonorm 8.1 (1991-2012) - Synthetic

System summary

Grid-Connected System

Sheds, single array

PV Field Orientation
Fixed plane
Tilt/Azimuth 20 / 0 °

Near Shadings
Linear shadings : Fast (table)

User's needs
Unlimited load (grid)

System information

PV Array

Nb. of modules	8840 units
Pnom total	4995 kWp

Inverters

Nb. of units	16 units
Pnom total	3999 kWac
Pnom ratio	1.249

Results summary

Produced Energy	7078053 kWh/year	Specific production	1417 kWh/kWp/year	Perf. Ratio PR	85.65 %

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General parameters

Grid-Connected System — **Sheds, single array**

PV Field Orientation

Orientation	
Fixed plane	
Tilt/Azimuth	20 / 0 °

Sheds configuration		
Nb. of sheds	17	units
Single array		
Sizes		
Sheds spacing	8.00	m
Collector width	4.58	m
Ground Cov. Ratio (GCR)	57.2	%
Top inactive band	0.02	m
Bottom inactive band	0.02	m
Shading limit angle		
Limit profile angle	23.1	°

Models used	
Transposition	Perez
Diffuse	Perez, Meteonorm
Circumsolar	separate

Horizon
Free Horizon

Near Shadings
Linear shadings : Fast (table)

User's needs
Unlimited load (grid)

Bifacial system
Model: 2D Calculation unlimited sheds

Bifacial model geometry		
Sheds spacing	8.00	m
Sheds width	4.62	m
Limit profile angle	23.1	°
GCR	57.7	%
Height above ground	1.00	m

Bifacial model definitions		
Ground albedo	0.25	
Bifaciality factor	80	%
Rear shading factor	5.0	%
Rear mismatch loss	10.0	%
Shed transparent fraction	0.0	%

PV Array Characteristics

PV module

Manufacturer	Waaree Energies	
Model	BiN-08-565	
(Custom parameters definition)		
Unit Nom. Power	565	Wp
Number of PV modules	8840	units
Nominal (STC)	4995	kWp
Modules	340 string x 26	In series
At operating cond. (50°C)		
Pmpp	4620	kWp
U mpp	1034	V
I mpp	4467	A

Inverter

Manufacturer	Sungrow	
Model	SG250-HX	
(Original PVsyst database)		
Unit Nom. Power	250	kWac
Number of inverters	192 * MPPT 8%	16 units
Total power	3999	kWac
Operating voltage	500-1450	V
Pnom ratio (DC:AC)	1.25	
No power sharing between MPPTs		

Total PV power

Nominal (STC)	4995	kWp
Total	8840	modules
Module area	22836	m²

Total inverter power

Total power	3999	kWac
Number of inverters	16	units
Pnom ratio	1.25	

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Array losses

Array Soiling Losses
Loss Fraction 3.0 %

Thermal Loss factor
Module temperature according to irradiance
Uc (const) 29.0 W/m²K
Uv (wind) 0.0 W/m²K/m/s

DC wiring losses
Global array res. 3.8 mΩ
Loss Fraction 1.5 % at STC

LID - Light Induced Degradation
Loss Fraction 0.5 %

Module Quality Loss
Loss Fraction 0.0 %

Module mismatch losses
Loss Fraction 2.0 % at MPP

Strings Mismatch loss
Loss Fraction 0.1 %

IAM loss factor
Incidence effect (IAM): Fresnel, AR coating, n(glass)=1.526, n(AR)=1.290

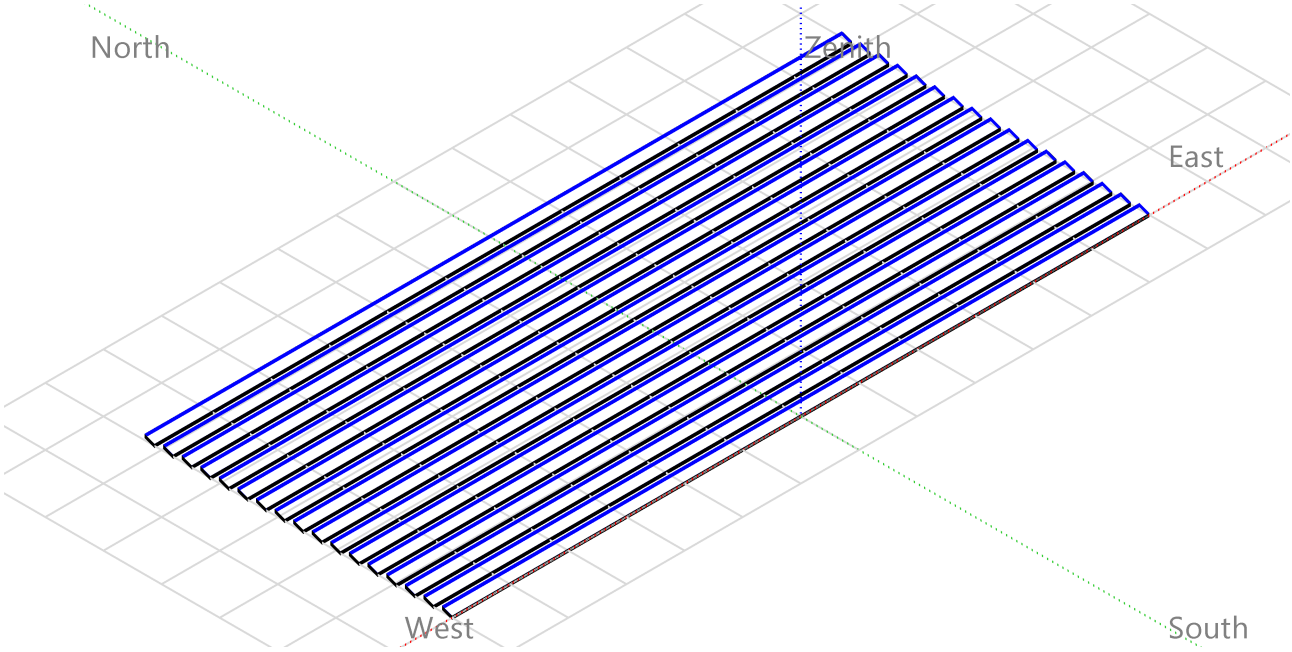
0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.999	0.987	0.962	0.892	0.816	0.681	0.440	0.000

System losses

Unavailability of the system
Time fraction 1.0 %
3.7 days,
3 periods

Near shadings parameter

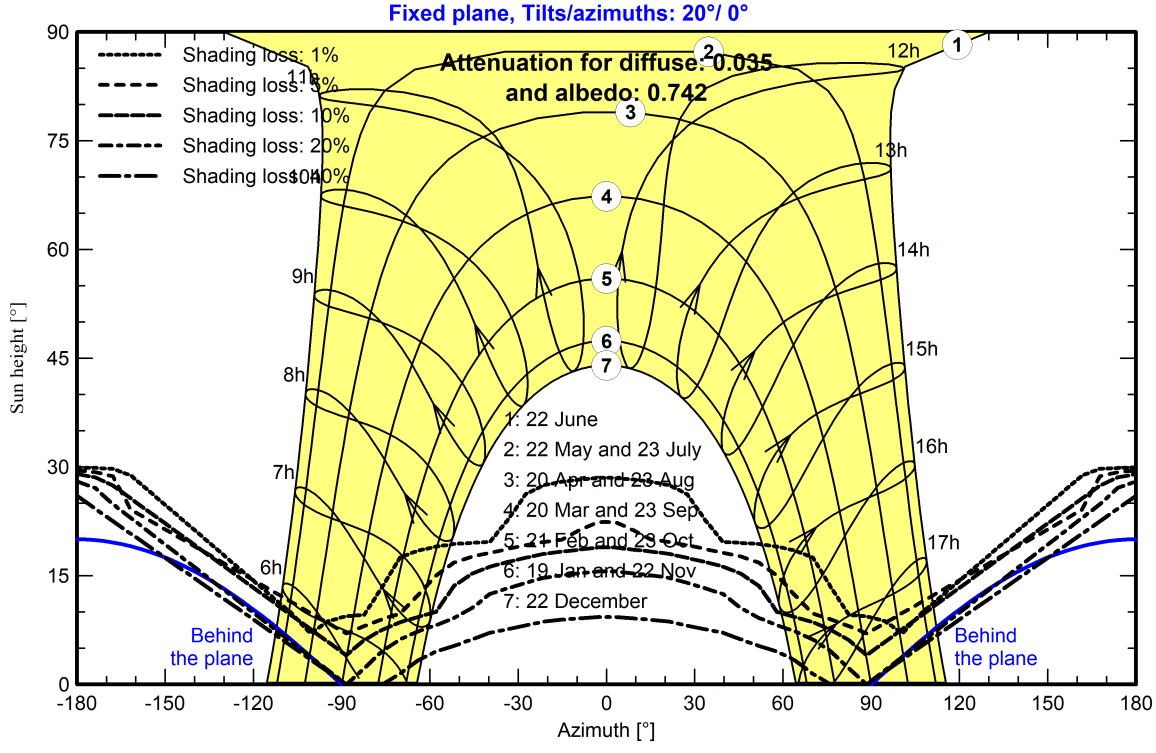
Perspective of the PV-field and surrounding shading scene



Iso-shadings diagram

Orientation #1

Fixed plane, Tilts/azimuths: 20°/ 0°

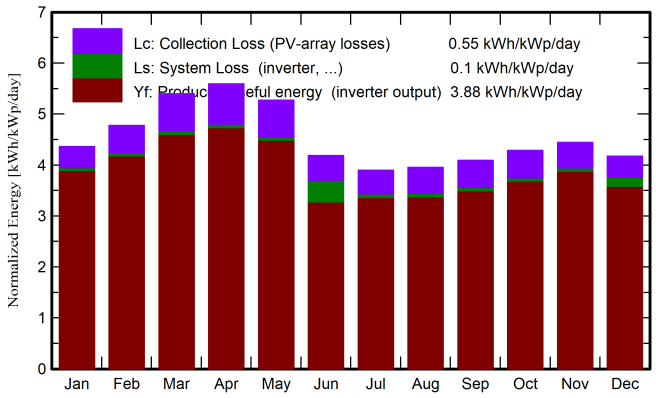


Main results

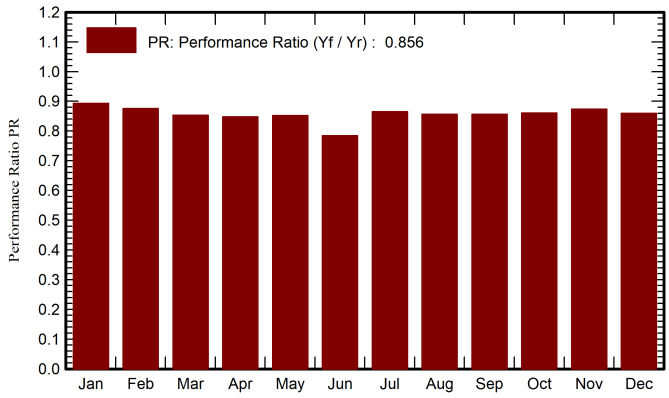
System Production

Produced Energy	7078053 kWh/year	Specific production	1417 kWh/kWp/year
		Perf. Ratio PR	85.65 %

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

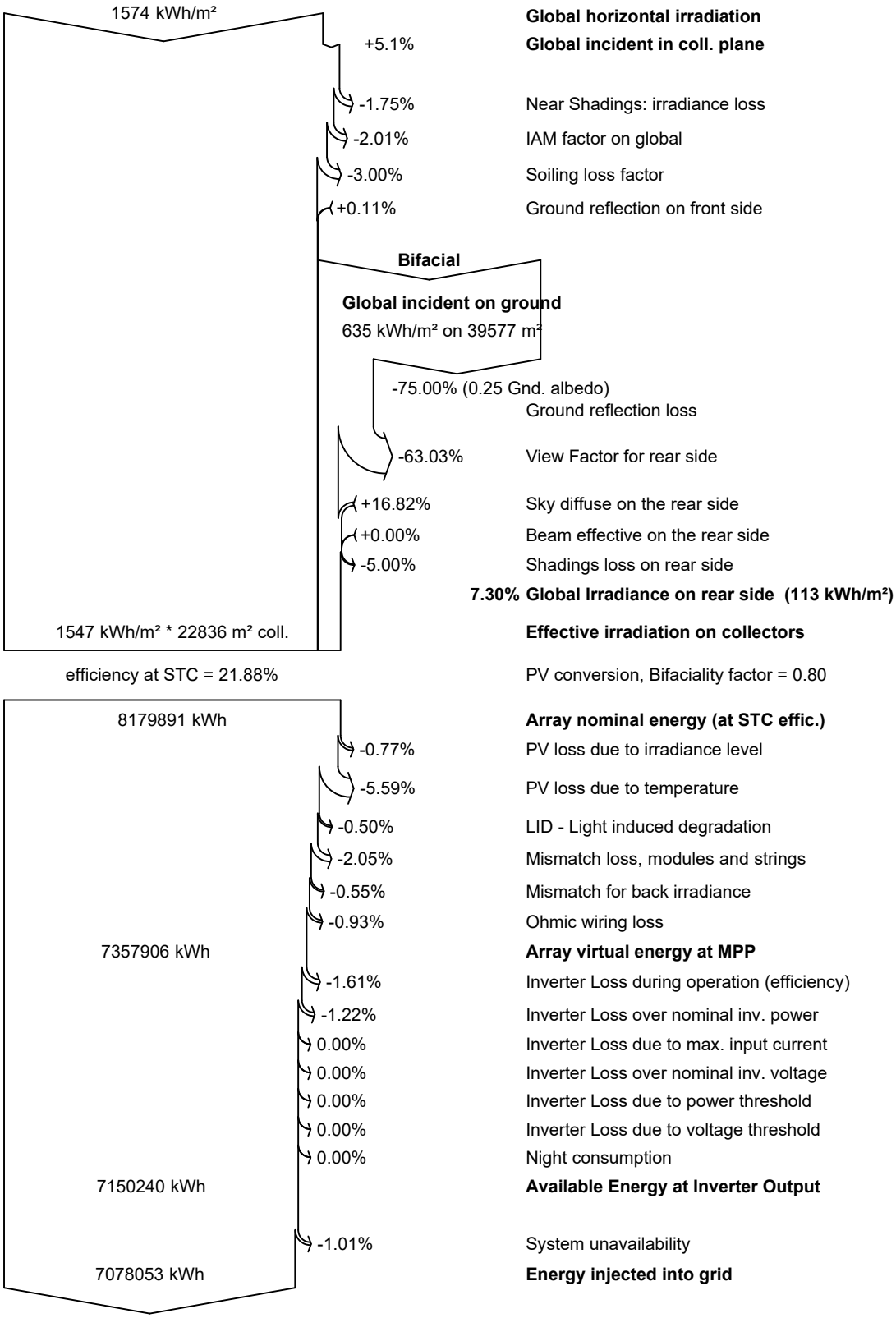
	GlobHor kWh/m²	DiffHor kWh/m²	T_Amb °C	GlobInc kWh/m²	GlobEff kWh/m²	EArray kWh	E_Grid kWh	PR ratio
January	112.1	60.55	17.45	135.2	126.7	611856	602898	0.893
February	118.0	67.00	22.10	133.7	125.5	593718	584978	0.876
March	156.9	85.04	27.55	167.3	157.1	723424	712927	0.853
April	166.5	88.89	30.18	167.7	157.4	720603	709930	0.847
May	171.7	96.70	31.43	163.5	152.8	706818	695431	0.852
June	134.0	93.63	30.47	125.5	116.3	555398	491910	0.785
July	128.3	87.28	29.78	120.8	111.9	532542	521809	0.865
August	125.0	83.59	29.36	122.5	113.8	534466	524042	0.857
September	120.4	78.22	28.31	122.8	114.3	534374	524913	0.856
October	121.8	73.59	27.25	132.9	124.3	581200	571700	0.861
November	112.8	60.86	23.36	133.4	125.2	591292	582257	0.874
December	106.1	57.07	18.68	129.3	121.4	583649	555258	0.860
Year	1573.6	932.42	26.34	1654.6	1546.9	7269341	7078053	0.856

Legends

GlobHor	Global horizontal irradiation
DiffHor	Horizontal diffuse irradiation
T_Amb	Ambient Temperature
GlobInc	Global incident in coll. plane
GlobEff	Effective Global, corr. for IAM and shadings
EArray	Effective energy at the output of the array
E_Grid	Energy injected into grid
PR	Performance Ratio

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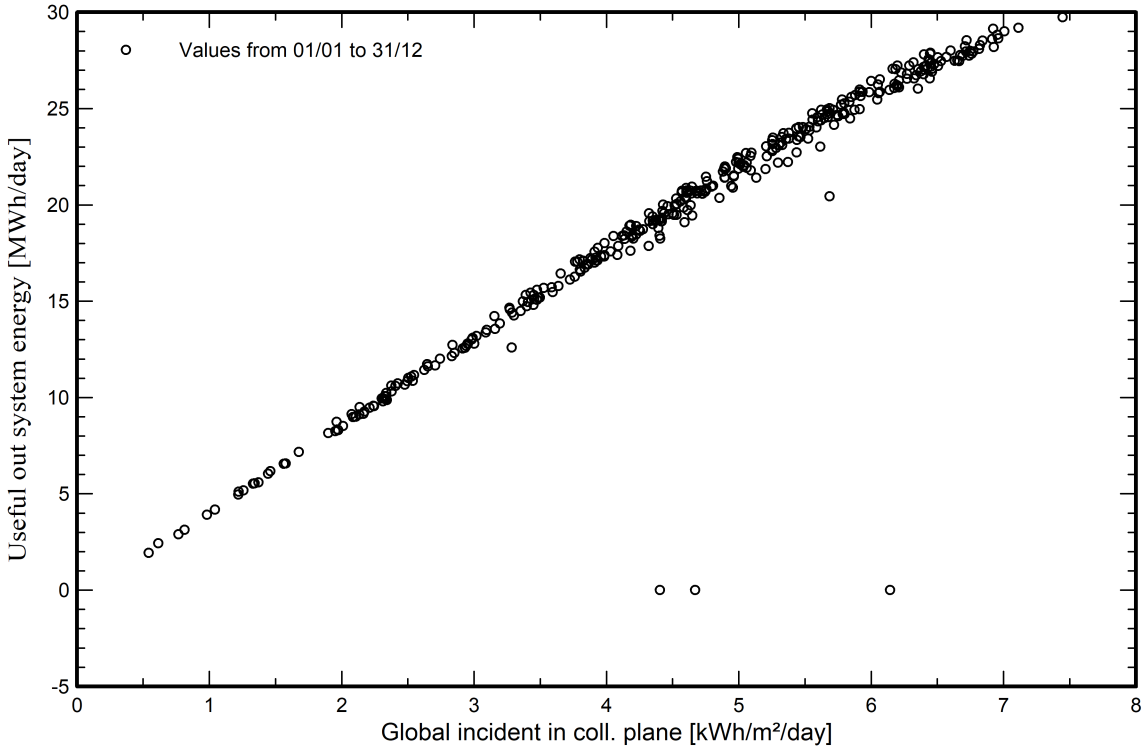
Loss diagram



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Predef. graphs

Daily Input/Output diagram



System Output Power Distribution

